

A Dual-Attention Mechanism Deep Learning Network for Mesoscale Eddy Detection by Mining Spatiotemporal Characteristics

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(Manuscript received 2 October 2021, in final form 18 April 2022)

ABSTRACT: Mesoscale eddies are a mechanism for ocean energy transfer, and identifying them on a global scale provides a means of exploring ocean mass and energy exchange between ocean basins. There are many widely used model-driven methods for detecting mesoscale eddies; however, these methods are not fully robust or generalizable. This study applies a data-driven method and proposes a mesoscale detection network based on the extraction of eddy-related spatiotemporal information from multisource remote sensing data. Focusing on the northwest Pacific, the study first analyzes mesoscale eddy characteristics using a combination of gridded data for the absolute dynamic topography (ADT), sea surface temperature (SST), and absolute geostrophic velocity (UVG). Then, a deep learning network with a dual-attention mechanism and a convolutional long short-term memory module is proposed, which can deeply exploit spatiotemporal feature relevance while encoding and decoding information in the gridded data. Based on the analysis of mesoscale eddy characteristics, ADT and UVG gridded data are selected to be the inputs for the detection network. The experiments show that the accuracy of the proposed network reaches 93.38%, and the weighted mean dice coefficient reaches 0.8918, which is a better score than those achieved by some of the detection networks proposed in previous studies, including U-Net, SymmetricNet, and ResU-Net. Moreover, compared with the model-driven approach used to generate the ground-truth dataset, the network method proposed here demonstrates better performance in detecting mesoscale eddies at smaller scales, partially addressing the problem of ghost eddies.

KEYWORDS: Deep learning; Eddies; Neural networks

1. Introduction

Mesoscale eddies, which are defined as eddies with a radius of tens to hundreds of kilometers and a duration of tens to hundreds of days, are common throughout the global ocean (Wyrki et al. 1976; Richardson 1983). Mesoscale eddies can be divided into cyclonic eddies (CEs) and anticyclonic eddies (AEs) according to their direction of rotation, and the different characteristics of AEs and CEs can be reflected in the direction of ocean currents and sea surface height and temperature anomalies. Mesoscale eddies have important impacts on ocean circulation, heat and salt transport, biological activity, and air–sea interactions in their surrounding areas (Chelton et al. 2011a). The northwest Pacific Ocean is one of the most active areas with respect to mesoscale eddies, and eddies are particularly prevalent in the Kuroshio–Oyashio Extension (KE) and Subtropical Countercurrent (STCC) regions, shown in Fig. 1. The flow between these two regions leads to significant spatiotemporal mesoscale eddy patterns (Yang et al. 2013). Mesoscale eddies in the northwest Pacific are therefore a popular focus of research, and this has resulted in some important findings that provide useful references for ocean circulation studies (Chen et al. 2014; Wang et al. 2020).

To date, several studies have investigated the geometric and physical characteristics of mesoscale eddies (Qiu and Chen 2010b; Chelton et al. 2011a). Based on these studies, researchers have developed many eddy detection methods that

facilitate in-depth investigation into the complex structure of mesoscale eddies and their impact on water flow (Nencioli et al. 2010; Mason et al. 2014; Ronneberger et al. 2015). These methods can be divided into two groups: model driven and data driven.

Model-driven methods are based on modeling the physical characteristics of mesoscale eddies. These methods can be divided into three categories: parameter based, geometric feature based, and contour based. The most representative physical parameter-based method is the Okubo–Weiss (OW) model (Okubo 1970; Weiss 1991), which has been used to detect eddies in the northeast Atlantic Ocean (Schütte et al. 2016), the northeast Atlantic Ocean (Schütte et al. 2016), and the Indian Ocean (Gulakaram et al. 2018; Albert and Bhaskaran 2020). The OW model uses different thresholds for different areas in different time intervals. This means that an appropriate threshold must be calculated in real time according to the local environment, leading to a weak generalization ability. The second category includes the winding-angle (WA) method (Sadarjoen et al. 1998; Portela 1997) and the vector geometric (VG) method (Nencioli et al. 2010). The WA method is based on Robinson's statement that a vortex exists when instantaneous streamlines mapped onto a plane normal to the vortex

Publisher's Note: This article was revised on 3 August 2022 to include funding information in the Acknowledgments that was omitted when originally published.

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DOI: 10.1175/JTECH-D-21-0128.1

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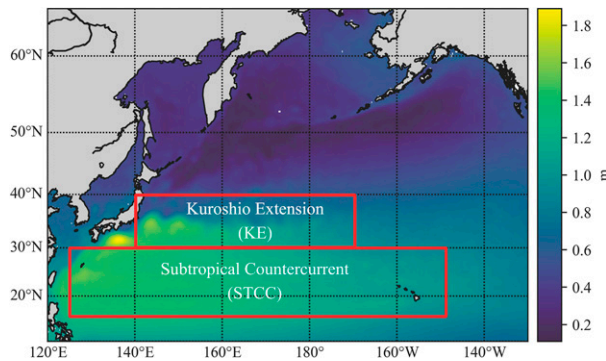


FIG. 1. Location of the KE (30°–40°N, 140°E–170°W) and STCC (20°–25°N, 125°E–155°W) regions, where mesoscale eddies occur frequently. Color map represents the root-mean-squared sea surface height variability.

core exhibit a roughly circular or spiral pattern (Robinson et al. 1984). The VG method was proposed by Nencioli et al. (2010) with four constraints concerning the eddy velocity field characteristics. The last category, the contour-based method, identifies mesoscale eddies by scanning gridded sea surface height (SSH) or sea level anomaly (SLA) data, searching for the local extremum, and drawing contours. Chelton et al. (2011b) proposed improvements to the standard contour-based method, and used a recursive contour searching algorithm that does not depend on a specific SSH threshold and enhances the generalization ability of the method. This improved contour-based method has been utilized to produce several frequently used mesoscale eddy datasets (Faghmous et al. 2012, 2015; Mason et al. 2014).

Each of the three model-driven detection methods mentioned above has specific strengths. The OW model can identify the grouping phenomenon in mesoscale eddies better than the other methods, whereas the WA and contour-based methods can better detect mesoscale eddies with waveform characteristics (Lian et al. 2019). To synthesize these strengths, some studies have adopted a hybrid identification method (Dong et al. 2011; Matsuoka et al. 2016), which has resulted in improved results. In summary, the model-driven methods described above have been verified and improved by many studies, but problems remain, such as low robustness and weak generalization ability (Li et al. 2020).

In recent years, an increasing number of observation satellites have provided shared oceanic data with wide spatial and temporal coverage, and some data-driven methods for mesoscale eddy detection have been proposed based on these data. Gridded oceanic remote sensing data such as SSH and SST can be understood as “images” that contain marine geographic information. Classification models including support vector machine (SVM) and random forest were applied to realize a binary classification that indicated whether there was a mesoscale eddy in the gridded data “images” (Ashkezari et al. 2016). Artificial neural network (ANN) was also used to judge the existence of mesoscale eddies from the gradient mask generated from remote sensing data (Castellani 2006, 2007). However, these classification models are not robust enough and cannot

realize large-scale mesoscale eddy detection. To further exploit the relationship between pixels, that is, to extract the spatial information of this “image,” a convolutional neural network (CNN) can be used. The idea behind a CNN is to process data in the form of an array or matrix (e.g., a two-dimensional image consisting of three color channels) using local connections, pooling, and multilayer stacking. The basic structure of a CNN includes the convolution layer, a pooling layer, a nonlinear layer, and a full connection layer (LeCun et al. 2015). With these operational layers, the spatial characteristics of the structured data can be learned from the simple to the deep.

CNNs have been widely used in image classification and image semantic segmentation, and have also played an important role in mesoscale eddy detection. Initially, basic CNN technology was applied to realize the binary classification of SSH images (Huang et al. 2017; Du et al. 2019) or high-resolution SST remote sensing gridded data (Moschos et al. 2020a,b). Later, networks with an encoder–decoder structure such as U-Net (Ronneberger et al. 2015), PSPNet (Zhao et al. 2017), and Deeplabv3+ (Chen et al. 2018) were used to perform semantic segmentation on gridded data, achieving a more direct, detailed, and extensive mesoscale eddy extraction (Franz et al. 2018; Lguensat et al. 2018; Santana et al. 2020; Xu et al. 2019). Most of these studies used single-source gridded SSH data as the input for the detection network, while some studies used a combination of multisource data as inputs (Fan and Zhong 2019; Santana et al. 2020). The technology of the proposal boundary box has also been used to determine the region of mesoscale eddies (Fan et al. 2020), following the basic principles of the RetinaNet (Lin et al. 2017) and Mask Region-CNN (Mask R-CNN) (He et al. 2017) methods.

According to previous studies, problems associated with model-driven methods, such as lost eddies, repeated eddies, and boundary eddies, can be alleviated by using a CNN approach (Xu et al. 2019). However, most currently available data-driven methods have failed to fully exploit spatiotemporal relationships within the data in terms of mesoscale eddy characteristics. A detection network can therefore be improved using prior knowledge of the spatiotemporal characteristics of mesoscale eddies. Some data may not be effectively related to mesoscale eddies, and should not be included in the inputs to a data-driven detection model. Thus, it is necessary to understand the basic characteristics of mesoscale eddies so as to select appropriate input data for the detection model. This prior knowledge can help us to choose appropriate data-driven methods, and to achieve better detection performance.

The purpose of this study is to accurately detect mesoscale eddies by extracting spatiotemporal features from a fusion of gridded oceanic remote sensing data. The study highlights the feasibility and superiority of data-driven methods for mesoscale eddy detection, and demonstrates the positive impact of fusing data from multiple sources. The key contributions presented in this paper are as follows.

- An image segmentation network is proposed to detect mesoscale eddies in the northwest Pacific. Following the general principles of ResU-Net, the network proposed in this paper consists of a dual-attention mechanism that uses

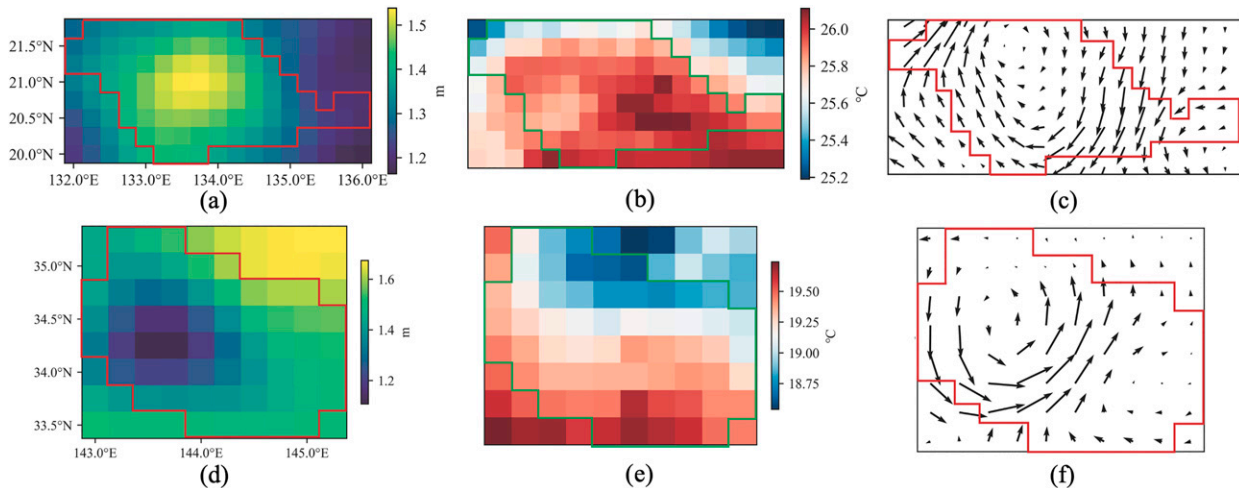


FIG. 2. Examples of mesoscale eddies (framed part) in the northwest Pacific between 2013 and 2016. (a) ADT for an AE. (b) SST for an AE. (c) UVG for an AE. (d) ADT for a CE. (e) SST for a CE. (f) UVG for a CE.

position attention (PA) and channel attention (CA) to explore the relevance of spatial traits.

- The network also implements a convolutional long short-term memory (ConvLSTM) module to extract spatiotemporal features. Experiments demonstrate that the proposed network results in good mesoscale eddy detection performance, with high accuracy and a high weighted mean dice coefficient.
- The results from different combinations of remote sensing gridded data are compared, and verify that absolute dynamic topography (ADT) and absolute geostrophic velocity (UVG) are most likely to provide the best combination of input data for the detection network.

2. Analysis and methodology

In this section, we first analyze the relationship between the gridded remote sensing data and the existence of mesoscale eddies. With the analysis results, a network for mesoscale eddies identification is proposed, which combines the prior knowledge of mesoscale eddy features for more effective detection.

a. Eddy traits analysis

The occurrence of mesoscale eddies can be reflected in changes to the marine physical environment, which can be monitored using specific remote sensing data. By analyzing these changes, we can learn about the spatial and temporal traits of mesoscale eddies. In this section, we analyze the spatiotemporal features of mesoscale eddies during 2013–16 in the northwest Pacific by combining three kinds of gridded remote sensing data: SSH, SST, and UVG. The SSH and UVG gridded data fields are provided by the Copernicus Marine Environment Monitoring Service (CMEMS), and SST data are provided by the Copernicus Climate Change Service (C3S). The extracted features will support the subsequent detection analysis.

1) RELATIONSHIPS BETWEEN EDDY TRAITS AND THE GRIDDED DATA

CEs in the Northern Hemisphere rotate counterclockwise, depressing the sea level at their center. Under geostrophic rotation, the seawater at the surface diverges and the cold-water mass from below rises. As a result, the SST and nutrient concentration decrease. AEs drive the opposite effects (Faghmous et al. 2012). Although some counterexamples are found in the North Pacific (Sun et al. 2019), these special eddies are not considered in this paper.

To analyze the mesoscale traits reflected in marine physical gridded data, we first apply the mesoscale eddy detection algorithm provided by Mason et al. (2014) to identify mesoscale eddies that occurred in a study region in the northwest Pacific (15°–39°N, 125°–173°E) during 2013–16. We then analyze the traits of the mesoscale eddies using the ADT, UVG, and SST data; UVG data are separated into zonal and meridional components. All of the datasets mentioned above are daily-mean data with a spatial resolution of $1/4^\circ \times 1/4^\circ$. Using the analysis described above, we check the status of ADT, SST, and UVG in the region that corresponds to the occurrence of mesoscale eddies. Examples for an AE and a CE on 1 January 2013, detected using the algorithm from Mason et al. (2014), are shown in Fig. 2. The figure shows that the AE is associated with a relatively high ADT and a high SST at its center, but that there is no clear local SST extremum. There is also a clockwise rotating vortex in the UVG field. Upon most occasions, these anomalies can be related to AE characteristics (McGillicuddy 2016). The CE is associated with the opposite anomalies to AE, and does not correspond to a clear local extremum in the SST field.

Zonal temperature profiles for an AE and a CE are shown in Fig. 3. As shown, there are local temperature extremums in the subsurface layer (200–300 m under the sea surface), and the polarities of the extremums are consistent with the characteristics of mesoscale eddies. The impact of the mesoscale

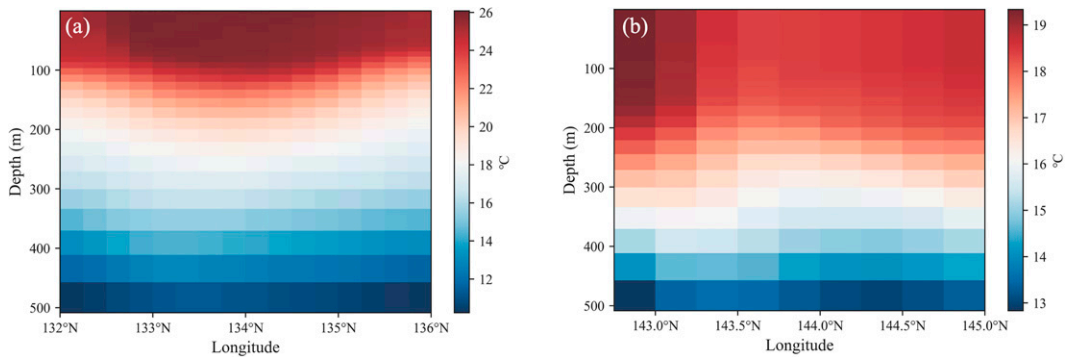


FIG. 3. Zonal temperature profiles that show how seawater temperature varies with depth in a mesoscale eddy. (a) AE (profile of 21°N). (b) CE (profile of 34.25°N).

eddies on seawater temperature in the subsurface layer is discernible. Moreover, the change of SST is a complex process that could be either because of the air–sea interaction or the mixing of seawater due to ocean dynamics (Hausmann et al. 2017). As a result, the dynamic effect of the mesoscale eddies is clear in the subsurface layer, whereas there is no clear corresponding extremum in the surface water layer. However, it is greatly difficult to obtain subsurface temperature data over a large range in real time. If subsurface temperature is added as a feature to the network for training, corresponding data are required when applying it. To ensure the generality of our network, we do not take subsurface temperature data into account.

Meanwhile, although high-resolution ($1 \text{ km} \times 1 \text{ km}$) SST gridded data have proven to be appropriate for mesoscale eddy detection, since these data reflect minor changes in the SST field (Moschos et al. 2020a,b), high-resolution data are not as accessible as the $1/4^\circ \times 1/4^\circ$ data and are incompatible with the resolution of the ADT and UVG data. Therefore, we also do not use high-resolution SST data.

In conclusion, the existence of mesoscale eddies is reflected in anomalies in the $1/4^\circ \times 1/4^\circ$ gridded ADT and UVG data, and the characteristics of the anomalies are different for different types of mesoscale eddies. The presence of mesoscale eddies also causes changes in sea temperature. However, the $1/4^\circ \times 1/4^\circ$ SST data within the eddies do not have an obvious extreme value, and their horizontal distribution is irregular. Thus, SST may be not a suitable input data for the mesoscale eddy detection network that we build next. The performance of different combinations of input data will be tested and discussed further in section 3.

2) SCALE AND SPATIAL TRAITS

The geometric scale and spatial distribution of mesoscale eddies in the study area is important for the design of the detection network structure and for the selection of the training parameters, and is particularly useful for exploiting relevant contextual data appropriately. Based on the detection principles of Chelton et al. (2011b) and Mason et al. (2014), the number of pixels within the eddy, I , should satisfy $5 \leq I \leq 1000$; the amplitude of the absolute difference between the extremum

of sea level height within the eddy and the average sea level height at the edge of the eddy, A , should satisfy $0.01 \leq A \leq 1.5 \text{ m}$; and there should be only one local extremum of sea level height within one mesoscale eddy. The center of the mesoscale eddy is defined as the location of the sea level height extremum, and the radius of the mesoscale eddy, r_E , is defined as the radius of a circle with a pixel area equal to the area covered by the mesoscale. The value of r_E can be calculated as follows:

$$r_E = \frac{2\pi R \cos \theta}{\omega} \sqrt{\frac{I}{\pi}} \quad (1)$$

where R is the radius of Earth, θ is the latitude of the eddy center location, and ω is the spatial resolution coefficient. Since the spatial resolution of the data used in this study is $1/4^\circ \times 1/4^\circ$, ω is set to $\omega = 360 \times 4 = 1400$, meaning that each line of latitude is divided evenly into 1400 segments.

To avoid the effect of intermittent eddies, only mesoscale eddies with $r_E \geq 30 \text{ km}$ and $A \geq 0.03 \text{ m}$ are considered. The statistical distributions of the radius and amplitudes calculated for the mesoscale eddies in the study area during 2013–16 are shown in Fig. 4. It can be seen that the radius distribution is similar to the Rayleigh distribution, and the majority of values fall in the range of 40–50 km. The number of CEs exceeds that of AEs for radius smaller than 80 km, while the opposite is true for eddies with radius larger than 80 km. The amplitude of most mesoscale eddies is in the range of 0–0.2 m, and the number of eddies in each bin decreases as the amplitude increases.

As shown in Fig. 5, mesoscale eddies are concentrated between 17° and 30°N . In this region, CEs occur mainly between 17° and 30°N , whereas the distribution of AEs is bimodal with a broad peak between 17° and 30°N and a narrow peak between 37° and 39°N . In summary, mesoscale eddies in the northwest Pacific occur most frequently in the STCC and KE regions due to the ocean system transitions (Qiu and Chen 2010c,a; Chang et al. 2015).

3) TEMPORAL TRAITS

As the development of mesoscale eddies is a continuous process, their temporal traits may provide information that is

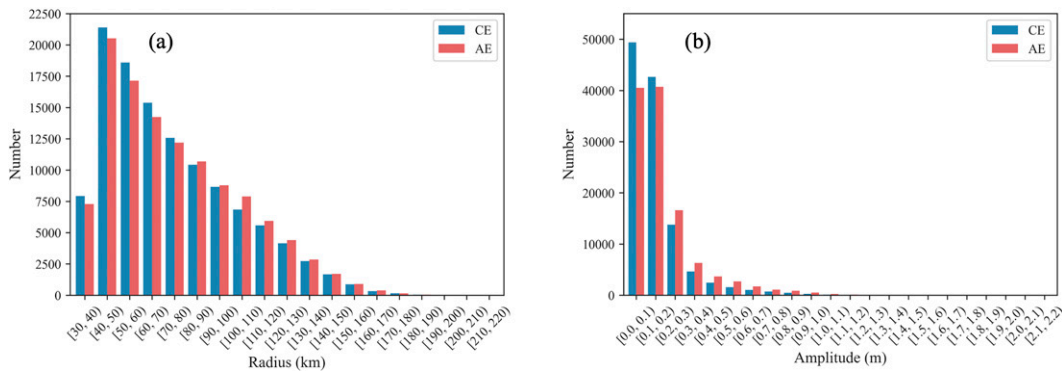


FIG. 4. Histograms of scales for mesoscale eddies in the study area between 2013 and 2016. (a) Radius. (b) Amplitude.

valuable for detection networks. Based on the widely used mesoscale eddy tracking algorithm proposed by Chelton et al. (2011b), we search for the subsequent eddy on the $(k + 1)$ th day within a circle of radius $\sqrt{2}^\circ$ (about 130 km) centered on the local ADT extreme of the mesoscale eddy found on the k th day. The closest mesoscale eddy found in this range is identified as the subsequent eddy on the $(k + 1)$ th day, and its moving trajectory within the life cycle is continuously extended. To reduce the influence of ghost eddies (eddies that are suddenly lost for some reason during their trajectory), a tracking lost buffer of one day is allowed.

There are 4320 intermittent mesoscale eddies with a recorded lifetime no more than 3 days in the original tracking results. To reduce the influence of intermittent eddies, only mesoscale eddy trajectories with a lifetime of more than 14 days are selected for analysis. A total of 4756 mesoscale eddy trajectories that meet the conditions are recorded in the study area for 2013 to 2016—2293 AE and 2463 CE trajectories—with lifetimes ranging from 14 to 397 days. Figure 6a shows the distribution of mesoscale eddy lifetimes. As shown, the lifetimes of most mesoscale eddies are concentrated in the range of 14–28 days, and the number of AEs with a longer lifetime (≥ 98 days) is more than that of CEs.

The change in position of the mesoscale eddy center during its life span is also analyzed. As shown by the zonal transport

in Fig. 6b, nearly 70% of mesoscale eddies tend to move westward during their lifetime in the record.

b. Method

The network we propose is based on ResU-Net, with some improvements added to meet the needs of our study. The network structure is shown in Fig. 7. In this section, we describe the details of the proposed network, including its overall structure and several core modules.

1) OVERALL NETWORK STRUCTURE

The essence of mesoscale eddy detection from gridded multisource remote sensing data is pixel-wise classification. The detection network should analyze spatiotemporal variations in the input data and map the extracted features to the occurrence of different types of mesoscale eddy. There are three main challenges. First, the gridded input data have a low spatial resolution, meaning that there is no clear boundary for target detection and segmentation. Second, although the occurrence of mesoscale eddies is reflected in the gridded ADT and UVG data, as shown in section 2, these two kinds of data have different traits in different eddy regions. Finally, both the spatial and temporal characteristics of the data should be considered, so as to achieve optimal detection.

To address these challenges, we propose a dual-attention mechanism deep-learning network. The network has a U-shaped encoding-decoding structure and a skip connection that connects the encoding and decoding parts with the same resolution level. This structure has been proven effective for image semantic segmentation, particularly in medical and remote sensing image segmentation (Lu and Tong 2019). The encoding and decoding parts have similar residual convolution block structures (ResBlock) (Santana et al. 2020) to ensure that the feature extraction and mapping are effective. The decoding part of the network contains extra PA and CA modules that make more efficient use of the features extracted by the encoding part and deal with the problem of multisource data fusion. The ConvLSTM is also implemented within the network to combine the spatial and temporal features.

The proposed network is shown in Fig. 7a. The encoding part handles feature extraction processes, in which information

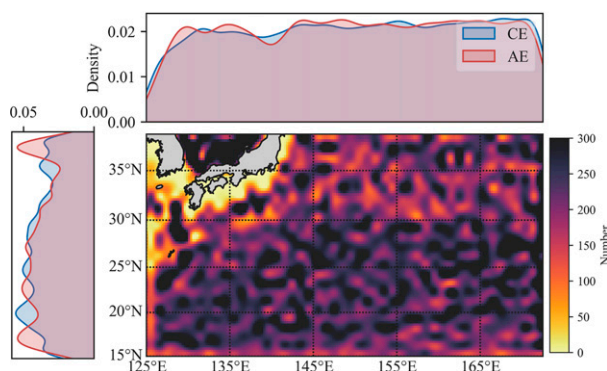


FIG. 5. Density of the mesoscale eddy distribution in the study area between 2013 and 2016.

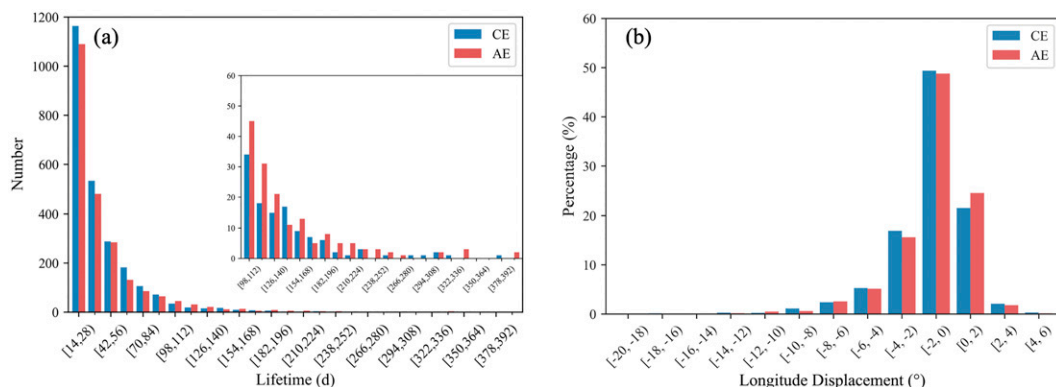


FIG. 6. Histograms of the temporal characteristics of mesoscale eddies in the study area between 2013 and 2016. (a) Mesoscale eddy lifetimes (partially zoomed). (b) Mesoscale eddy zonal movement. A negative change in longitude indicates westward movement, whereas a positive change indicates eastward movement.

is condensed and abstracted. The decoding part handles feature mapping processes, transforming the extracted features into the required classification results. Some information loss is inevitable during these processes. This problem is partially addressed by the skip connection, which transfers information directly from the encoding part (Ronneberger et al. 2015). Overall, the structure of the network makes it suitable for detecting targets like mesoscale eddies in remote sensing images, since the data are complex, with a large range of gray fuzzy boundaries and poor semantic information.

2) RESIDUAL CONVOLUTION BLOCK

The ResBlock is designed to deepen the network while alleviating the problem of network degradation (He et al. 2016), as shown in Fig. 7b. The input to the ResBlock, x_r , is processed in two ways. A 1×1 convolution is used to obtain a direct linear mapping result; i.e., $W_r * x_r$, where W_r denotes the convolutional filter. Meanwhile, x_r is subjected to the following processes

twice in sequence: batch normalization (BN), a rectified linear unit (ReLU), and the two-dimensional convolution (Conv2D). After the repeated processes, a nonlinear residual mapping result is generated: $\mathcal{F}(x_r)$. For the l th ResBlock, the output $\mathcal{H}(x_r)$ can be recursively described as follows:

$$x_r^l = \mathcal{H}(x_r^{l-1}) = x_r^1 \prod_{s=1}^{l-1} W_r^s + \sum_{u=1}^{l-1} \left(\prod_{v=u+1}^{l-1} W_r^v \right) \mathcal{F}(x_r^u), \quad (2)$$

where x_r^1 is the original input to the first ResBlock. x_r^l and W_r^l are the input and the 1×1 convolution filter for the l th ResBlock, respectively.

By adding the direct and residual mapping, the ResBlock combines deep-learning and shallow-learning features, meaning that it can extract more valuable information. The original information is maintained and passed by the process of linear mapping with a 1×1 convolution, which reduces the possibility of degradation.

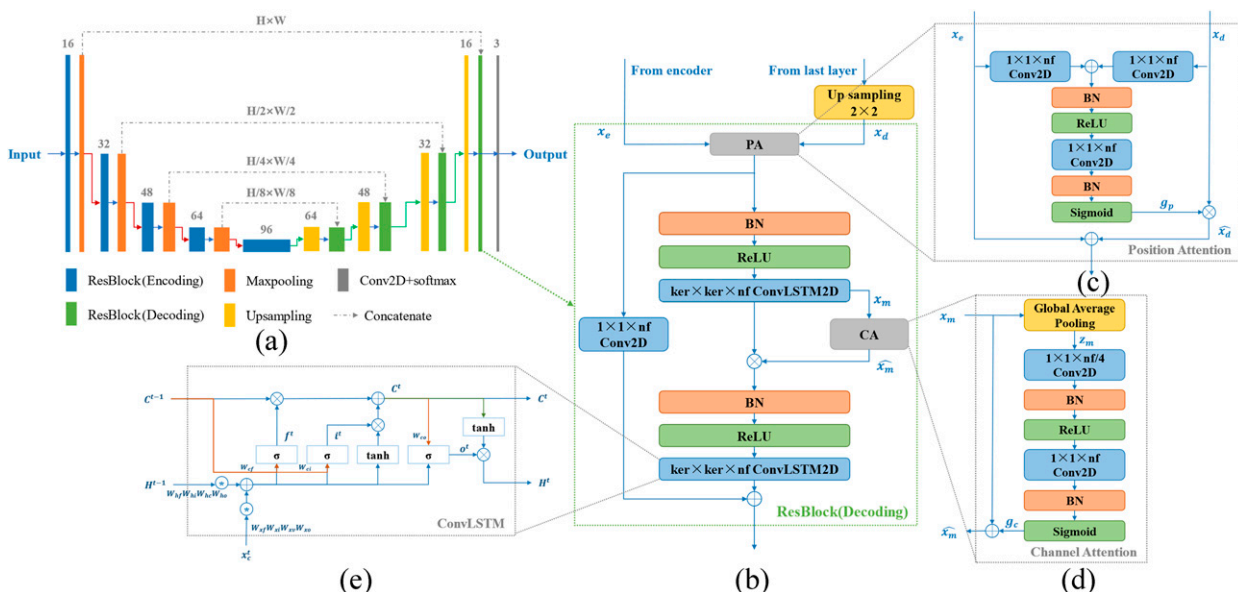


FIG. 7. Structure of the detection network. (a) The overall detection network. (b) The ResBlock in the decoding part of the network. (c) The PA module. (d) The CA module. (e) The ConvLSTM module.

3) PA MODULE

The study area contains part of the nonmarine region where mesoscale eddies cannot exist. Moreover, previous studies and our analysis in [section 2](#) show that mesoscale eddies have a distribution peak in certain regions. The PA module is therefore used to emphasize areas where mesoscale eddies occur frequently and to ignore nonmarine regions, resulting in improved detection.

The PA module is implemented before each ResBlock in the decoding part of the network, as shown in [Fig. 7c](#). First, the feature x_e , which is conveyed from the encoding part by skip connection, and feature x_d , which results from the previous up-sampling layer of the decoding part, go through two 1×1 convolutional layers separately. The first convolutional layer is followed by a ReLU whereas the second is followed by a sigmoid layer, which allows the position weight matrix, g_p , to be calculated. By multiplying x_d with g_p , position attention weighting is implemented to $\hat{x}_d$. Finally, the weighted feature $\hat{x}_d$, integrated with x_e , is sent to the ResBlock module. The process can be expressed as follows:

$$g_p = \delta[W_p * \sigma(W_e * x_e + W_d * x_d)], \quad (3)$$

$$\hat{x}_d = x_d \cdot g_p, \quad (4)$$

where the asterisk (*) denotes a convolutional process, and the dot (·) denotes an element-wise product. W_p , W_e , and W_d are the convolutional filters for x_e , x_d , and the process for generating g_p , respectively. After training, the PA module increases the weight in the regions where eddies occur frequently. In this way, more eddies are detected in key areas such as the KE and STCC regions, improving the accuracy of the detection network.

4) CA MODULE

The analysis in [section 2](#) confirms that the presence of mesoscale eddies is clearly related to ADT and UVG anomalies. One existing challenge associated with fusing data from these different data is that the encoder may extract different features, which may contribute differently to the mesoscale eddy detection. In addition, in the processes of feature extraction and up-down sampling, relevant semantic environment may be destroyed as the receptive field changes. As a result, the correlations between different features from different sources may not be fully utilized.

The CA module is added to the ResBlock of the decoding part of the network to address this problem, as shown in [Fig. 7d](#). Within the ResBlock, the result, x_m , generated from the first convolutional layer and the activation layer, goes through a global average pooling layer first. Each channel of the feature atlas is replaced with its average value to condense the original information. For the n th channel of x_m , this process can be described as follows:

$$z_m^n = \frac{1}{H \times B} \sum_{a=1}^H \sum_{b=1}^B x_m^n(a, b), \quad (5)$$

where $x_m^n \in \mathbb{R}^{H \times B \times \text{nf}}$ is the n th channel of x_m ; $z_m^n \in \mathbb{R}^{1 \times 1 \times \text{nf}}$ is the n th channel of z_m , which is the global average pooling

result for x_m ; H and B denote the height and breadth of the feature maps; and nf denotes the number of channels.

The condensed information z_m goes through a 1×1 convolutional layer to reduce the number of dimensions. Following a BN layer and a ReLU layer, there is another 1×1 convolutional layer that can restore the number of channels. The relationship among all the extracted features can be learned during these condensing and restoring processes ([Hu et al. 2018](#)). Sigmoid activation is then used to obtain the attention weight for each channel, g_c . Finally, g_c is applied to x_m , generating a weighted feature $\hat{x}_m$. The calculating process is similar to Eqs. (3) and (4). In this way, attention can be imposed on each channel and the important feature can be emphasized, leading to more effective detection.

5) CONVLSTM MODULE

In addition to scale and spatial characteristics, such as radius, amplitude, and location distribution, the analysis in [section 2](#) shows that the temporal characteristics of mesoscale eddies are also clearly discernible in gridded remote sensing data. These temporal characteristics provide a further important reference for mesoscale eddy detection. For example, the distribution of mesoscale eddies on the k th day can be roughly inferred from the distribution of mesoscale eddies on the $(k - 2)$ th and $(k - 1)$ th days, using the lifetime of mesoscale eddies and their zonal movement trends. These temporal characteristics are not spatially independent; thus, spatial traits and temporal traits should be dealt with together.

A ConvLSTM module is used in the decoding part of the network to consider temporal traits. The distinguishing characteristic of the ConvLSTM module is that it combines the advantages of CNN and long short-term memory (LSTM) to realize the extraction, exploiting spatial and temporal features. The ConvLSTM module is shown in [Fig. 7e](#), where x_c^t is the input to the module at the moment t ; C^t and H^t represent the module's long-term memory and short-term memory at moment t , respectively; and f^t , i^t , and o^t are the forget, input, and output gates at moment t , respectively. The difference between ConvLSTM and the original LSTM modules is that all variables in the ConvLSTM module are three-dimensional tensors with a scale of $H \times B \times \text{nf}$, and convolution operations are added. The calculations are as follows:

$$i^t = \sigma(W_{xi} * x_c^t + W_{hi} * H^{t-1} + W_{ci} \cdot C^{t-1} + b_i), \quad (6)$$

$$f^t = \sigma(W_{xf} * x_c^t + W_{hf} * H^{t-1} + W_{cf} \cdot C^{t-1} + b_f), \quad (7)$$

$$C^t = f^t \cdot C^{t-1} + i^t \cdot \tanh(W_{xc} * x_c^t + W_{hc} * H^{t-1} + b_c), \quad (8)$$

$$o^t = \sigma(W_{xo} * x_c^t + W_{ho} * H^{t-1} + W_{co} \cdot C^t + b_o), \quad (9)$$

$$H^t = o^t \cdot \tanh(C^t), \quad (10)$$

where W_{xi} , W_{hi} , and W_{ci} denote filters used in the linear operation with x_c^t , H^{t-1} , and C^{t-1} at the input gate, respectively, and b_i , b_f , b_c , and b_o denote biases in each gate operation.

TABLE 1. Experiment results of previously proposed U-Net-like networks for the test dataset. The two boldface values show the highest average accuracy and average WMDC among the four different combinations of network and input data.

Network	Input data	Average accuracy	Average WMDC
U-Net	ADT	0.9068(± 0.0014)	0.8489(± 0.0028)
SymmetricNet	ADT + SST + UVG	0.8863(± 0.0008)	0.8169(± 0.0024)
ResU-Net	ADT	0.9073(± 0.0009)	0.8516(± 0.0026)
	ADT + SST + UVG	0.9153(± 0.0005)	0.8622(± 0.0020)

To cooperate with the ConvLSTM module, the input is sorted in a five-dimensional array with a shape of (*batch size*, *time*, *height*, *breadth*, *channel*). In the other non-ConvLSTM layers in the network, the data for each time frame are encapsulated for unified processing. Except for the last ConvLSTM layer in the decoding part, which outputs nonsequential data (i.e., time = 1), the ConvLSTM layers all output five-dimensional arrays with the same time length as the input data. In this way, the spatial and temporal characteristics of the mesoscale eddies can be combined, and mesoscale eddies from the last few days can be more closely associated, resulting in better detection performance.

3. Experiment and discussion

In this section, we preprocess the data according to the demands of the detection task, and conduct a series of experiments to compare the performance between different network structures and different sets of input data. To highlight the feasibility and advantages of the proposed detection network, we also demonstrate the positive impact of combining data from multiple sources, and present a comparison between detection results calculated by a model-driven method and by CNN-based methods.

a. Data collection and preprocessing

The input data for the detection network include ADT and UVG data provided by CMEMS, and SST data provided by C3S. All data were daily average data with a spatial resolution of $1/4^\circ \times 1/4^\circ$. It should be noted that UVG is divided into zonal and meridional geostrophic flow velocities, so the integrated dataset is actually a four-channel dataset. Data in the range of 15° – 39° N, 125° – 173° E from 1 January 1993 to 31 December 2016 are selected.

The data are provided in separate datasets; thus, the ADT and UVG data are used as references to complement the SST data by filling in missing values. Since the mesoscale eddy detection problem is similar to the natural image segmentation problem, the different data sources are normalized and unified to form grayscale images. Nonmarine areas are set to 255 to distinguish these from ocean areas. Data from 1993 to 2012 are used as the training dataset, with a shape of (7305, 96, 192, 4), and the remaining data are the test dataset, with a shape of (1461, 96, 192, 4).

TABLE 2. Experiment results for multisource data combinations for the test dataset using ResU-Net. The first two boldface values show the highest average accuracy and average WMDC among the three single-source input data, and the second two boldface values show the highest average accuracy and average WMDC among the four multisource input data.

Network	Input data	Average accuracy	Average WMDC
ResU-Net	ADT	0.9073(± 0.0009)	0.8516(± 0.0026)
	SST	0.6524(± 0.0092)	0.4658(± 0.0188)
	UVG	0.9158(± 0.0008)	0.8630(± 0.0024)
	ADT + SST	0.9130(± 0.0023)	0.8593(± 0.0035)
	ADT + UVG	0.9161(± 0.0010)	0.8632(± 0.0024)
	SST + UVG	0.9155(± 0.0011)	0.8626(± 0.0022)
	ADT + SST + UVG	0.9153(± 0.0005)	0.8622(± 0.0020)

The mesoscale eddy detection algorithm provided by [Mason et al. \(2014\)](#) is used for generating ground-truth datasets. The minimum and maximum areas for an identified mesoscale eddy are set to 5 and 1000 pixels, respectively. All pixels that correspond to AEs are assigned a value of -1 , and all pixels that correspond to CEs are assigned a value of 1 ; all other pixels are assigned a value of 0 . Images of these assigned values are similarly separated into a training dataset and test dataset.

b. Experiment design

To better compare the effectiveness of different networks using different combinations of input data, the parameters that may affect the training result should be made uniform. The upper and lower sampling depths of each network are set as 4, which has been proven in parameter adjustment experiments to be a relatively appropriate depth for accuracy and efficiency ([Ronneberger et al. 2015](#)). Since the mesoscale eddy semantic information is not overly complex, the number of channels is compressed at each layer. The numbers of channels at each layer, including in the transition between the encoding and decoding parts, are 16, 32, 48, 64, and 96. The size of convolution kernel is set as 3. To avoid overfitting, L2

TABLE 3. Experiment results of multisource data combinations for the test dataset using previously proposed networks. The two boldface values show the highest average accuracy and average WMDC among the eight different combinations of network and input data.

Network	Input data	Average accuracy	Average WMDC
ResU-Net	UVG	0.9158(± 0.0008)	0.8630(± 0.0024)
	ADT + UVG	0.9161(± 0.0010)	0.8632(± 0.0024)
ResU-Net + PA	UVG	0.9207(± 0.0010)	0.8705(± 0.0018)
	ADT + UVG	0.9223(± 0.0011)	0.8731(± 0.0021)
ResU-Net + PA + CA	UVG	0.9220(± 0.0007)	0.8726(± 0.0021)
	ADT + UVG	0.9225(± 0.0008)	0.8735(± 0.0018)
ResU-Net + PA + CA + ConvLSTM	UVG	0.9329(± 0.0039)	0.8903(± 0.0061)
	ADT + UVG	0.9338(± 0.0032)	0.8918(± 0.0053)

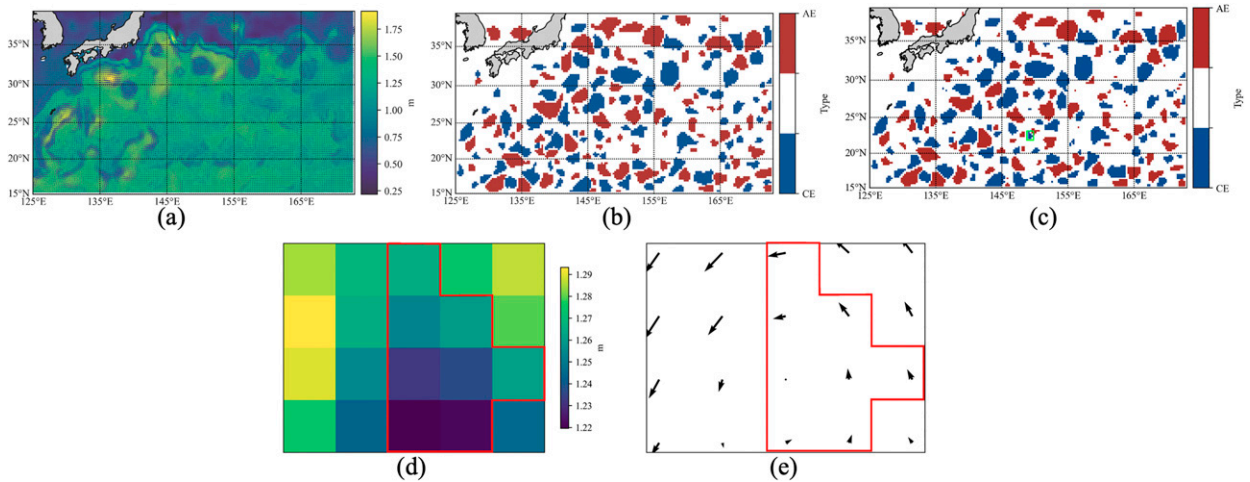


FIG. 8. Examples of eddy detection in the northwest Pacific between 2013 and 2016. (a) ADT and UVG gridded data for a single day. (b) Detection results using the ground-truth dataset. (c) Detection results for the proposed network with combined ADT and UVG data as input. The area framed by a green box is a small eddy, newly detected by the proposed network. (d) ADT for the newly detected eddy. (e) UVG for the newly detected eddy.

regularization of 0.001 is applied at each Conv2D/ConvLSTM layer. Meanwhile, the BN and dropout layers are added after each Conv2D/ConvLSTM layer. All of the networks are trained to meet the early stopping condition. A sixfold cross validation is carried out, and the average from these for the test dataset is taken as the final result. The experiment environment is based on one RTX-2080Ti GPU.

The detection of mesoscale eddies is essentially a pixel-level ternary classification problem. Categorical accuracy is the proportion of the total number of pixels that are correctly classified, expressed as a percentage, and is commonly used as an evaluation index for such problems. However, because the mesoscale eddy dataset is unbalanced, pixels of noneddy areas accounting for only 10% while pixels of AE and CE taking up nearly 45%, respectively, the categorical accuracy does not reflect the detection performance. Therefore, the weighted mean dice coefficient (WMDC) (Santana

et al. 2020) is introduced as a more appropriate evaluation index for this problem. The dice coefficient (DC) is calculated as follows:

$$DC = \frac{2|P \cap G|}{|P| + |G|}, \quad (11)$$

where P is the predicted result and G is the ground-truth value. The DC is calculated for each category, and the category DCs are weighted, according to the proportion of the total number of pixels that are accounted for by that category, to get the WMDC. The higher the WMDC, the better the detection performance.

c. Performance of networks

Several kinds of U-Net networks that have been applied to the problem of mesoscale detection are first tested and the

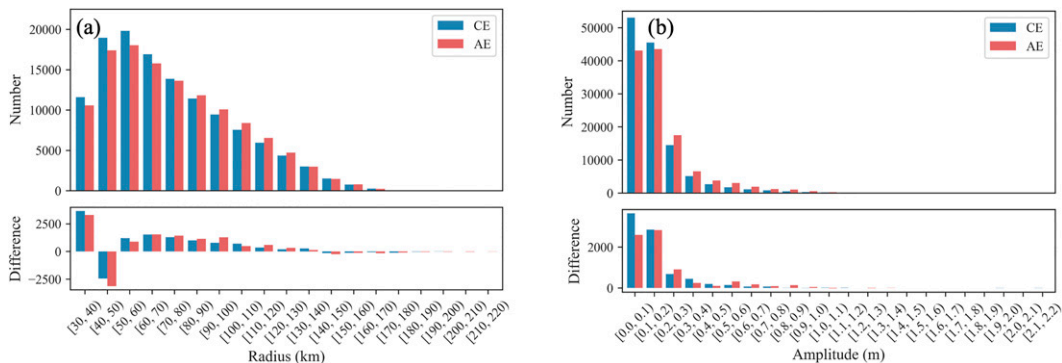


FIG. 9. (top) Histogram of scales for the mesoscale eddies detected by our network in the study area between 2013 and 2016. (bottom) Graphs indicating the difference between the number in each bin using the network-detected result, and using the ground-truth dataset. (a) Radius. (b) Amplitude.

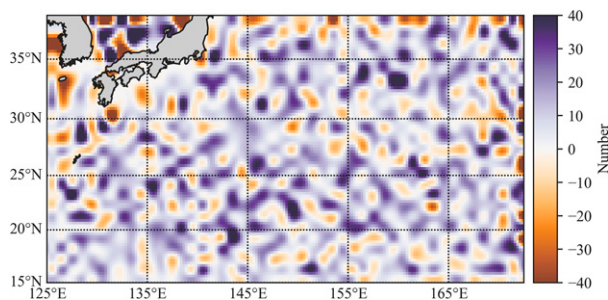


FIG. 10. Density map of the difference between the distribution of mesoscale eddies detected by our network and in the ground-truth dataset in the study area between 2013 and 2016.

results are shown in Table 1. As shown, the effectiveness of the detection network using ResU-Net is significantly higher than that of the other two networks without a ResBlock structure, and the detection networks that use multisource data as input perform better than those that use a single source of input data. The ResU-Net structure is therefore used as the benchmark against which the performance of each improved detection network is compared.

A series of experiments are carried out using ResU-Net to compare the effects of using different combinations of multisource data as inputs. It is worthy to note that UVG can be derived from ADT using mathematical model (Chu 2020). Therefore, using ADT as network input can indirectly include UVG information once the network successfully learns the relationship between ADT and UVG. However, the results generated by the mathematical model should be more accurate than those of machine learning. Hence, UVG data, which are an important component for mesoscale eddy detection besides ADT, are still included as network input. In this way, the model-driven and data-driven methods can supplement each other.

The results in Table 2 show that, when only single-source input data are used, the best detection performance is achieved using UVG data and the poorest results are achieved if SST is used, in terms of accuracy and WMDC. Where multiple data sources are combined as inputs, the combination that gives the best performance is ADT and UVG, whereas including SST

as a third source does not improve the detection performance. This confirms the conclusion presented in section 2. Therefore, in the subsequent improved network experiments, only the performance of detection network that use UVG, or ADT and UVG combined, are tested. Experiments with two kinds of input data are conducted using the improved network, and the experiment results are shown in Table 3. The results prove that including the PA, CA, and ConvLSTM modules makes the detection more effective, and the improvement is greatest when all three modules are included: the accuracy increases from 91.58% to 93.29%, and the WMDC increases from 0.8630 to 0.8903. Meanwhile, combining ADT and UVG as inputs to the network performs better in each experiment, proving that the combination of model-driven method and data-driven method does works. As a result, the final accuracy is 93.38% and the WMDC is 0.8918. Admittedly, networks with ConvLSTM modules take nearly 12 times longer than those without ConvLSTM modules to train, but the time consumed for testing the 4-yr dataset is still acceptable (2 s for networks without ConvLSTM modules and 14 s for networks with ConvLSTM modules). Additionally, there have been attempts to add the functional modules including PA, CA, and ConvLSTM to the encoding part, but the result shows little improvement, only adding unnecessary complexity. Therefore, the functional modules are only implemented to the decoding part.

d. Detection results

Our proposed network with combined ADT and UVG data as input is used for detection on testing data. The result is shown in Fig. 8, while the histograms for eddy radius and amplitude are shown in Fig. 9. To compare with the ground-truth dataset, only mesoscale eddies with $r_E \geq 30$ km and $A \geq 0.03$ m are considered. The result shows that the number of mesoscale eddies detected by our network is higher than that in the ground-truth dataset, and the average radius and amplitude are similar to those of the ground-truth dataset. The distributions for the network-detected eddies have a similar shape to those for the ground-truth dataset, but it is clear that our network detects more eddies with a relatively small radius (30–40 km) and small amplitude (≤ 0.2 m). Twenty newly detected mesoscale

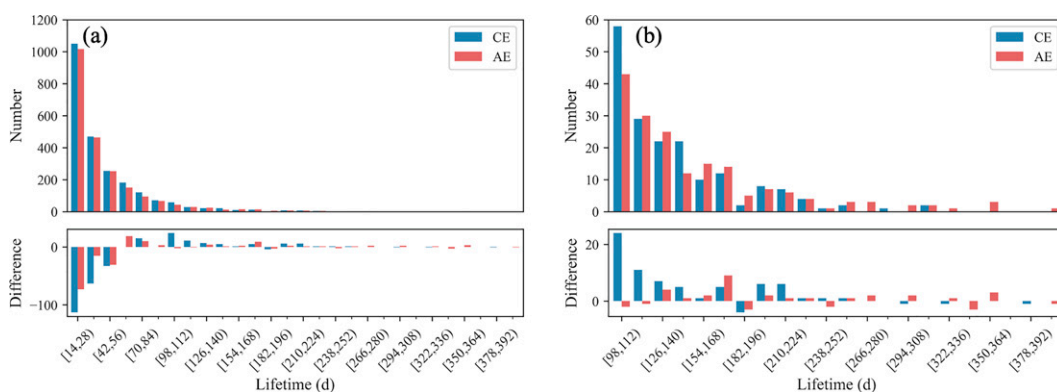


FIG. 11. Histograms of mesoscale eddy zonal movement trends in the study area between 2013 and 2016. (a) Overall. (b) Partially zoomed.

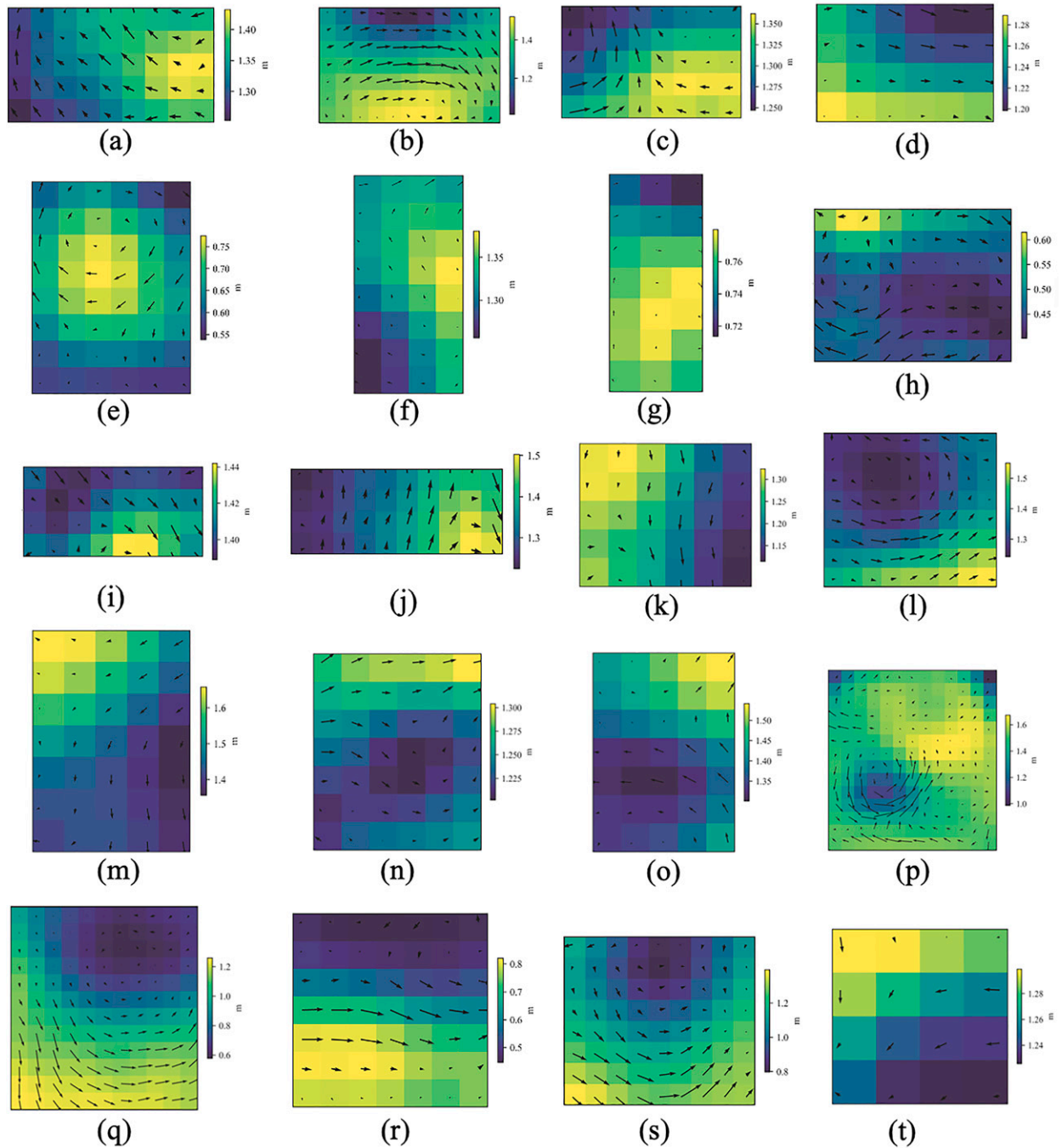


FIG. A1. Examples of some newly detected mesoscale eddies that are randomly selected from the network output. For each example, the color denotes ADT. (a)–(j) AE. (k)–(t) CE.

eddies (10 AEs and 10 CEs, respectively) are randomly selected and checked from the detection results. Nearly 75% of these selected mesoscale eddies are proved to be correct, showing corresponding characteristics in ADT and UVG data fields (the results are shown in the appendix).

The difference between the spatial distribution of eddies detected by our network and in the ground-truth dataset is

shown in Fig. 10. More eddies are detected in most areas within the study region. The mesoscale eddy trajectories with a lifetime of more than 14 days are analyzed to assess temporal traits. There are a total of 4556 mesoscale trajectories in the detection results generated by our network. As more eddies are successfully detected, more broken trajectories are linked. Therefore, the mesoscale eddy tracking algorithm can record more

trajectories that have a relatively long life, as shown in Fig. 11, making the mesoscale eddy tracking results more complete. The longest detected eddy lifetime reaches 725 days, whereas that in the ground-truth dataset is only 397 days. Since some of the broken trajectories are connected by the newly detected eddies, the total number of recorded trajectories declines slightly.

In summary, more smaller-scale mesoscale eddies can be detected in the northwest Pacific during 2013–16 using the proposed network with ADT and UVG data as inputs, particularly in the STCC and KE regions where mesoscale eddies occur frequently. The phenomenon of ghost eddies is somewhat alleviated using the proposed network since more eddies are detected successfully, leading to trajectory records with a longer lifetime.

4. Conclusions

In this paper, a deep-learning network with a dual-attention mechanism is proposed to detect mesoscale eddies. The network has an encoding–decoding structure with functional modules that include ResBlock, PA, CA, and ConvLSTM. It is able to exploit spatiotemporal features extracted from multisource remote sensing data, and to make full use of these features to map the occurrence of mesoscale eddies. The network is applied to detect mesoscale eddies in the northwest Pacific during 1993–2016, using gridded ADT and UVG data as inputs. Our experiments show that multisource inputs have a positive effect on detection performance. Compared with the model-driven method used to calculate the ground-truth dataset, the proposed network performs better for detecting smaller eddies and somewhat alleviates the problem of ghost eddies.

This study provides a strong basis for further research into mesoscale eddies. In future work, global mesoscale eddy data will be taken into consideration to broaden the range of applications for the detection network. We will further improve the network by introducing modules such as a temporal attention mechanism to achieve better detection results.

Acknowledgments. This Research is supported in part by the Zhejiang Provincial Natural Science Foundation under Grant LR19F030002, the Key Research and Development Program of Shandong Province under Grant 2019TSLH0203, and the Zhoushan City-Zhejiang University Joint Specific Project under Grant 2020C81002.

Data availability statement. The ADT and UVG gridded remote data can be downloaded from CMEMS (<http://marine.copernicus.eu>). The SST gridded remote data can be downloaded from C3S (<https://cds.climate.copernicus.eu/>). The open-source code of mesoscale eddy detection is provided by Mason et al. (2014) and Lguensat et al. (2018) (<https://github.com/AntSimi/py-eddy-tracker>).

APPENDIX

Visualization of the Newly Detected Eddies

Figure A1 presents examples of some newly detected mesoscale eddies that are randomly selected from the network output.

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